

Unit-II

Cognitive Models of memory & language:

⇒ It explains how human store, process, & use information & language.

- These models are based on cognitive science, psychology & AI.

- These helps in understanding human thinking.

(i) Cognitive Models of memory:

⇒ These models describe how info is encoded,

stored & retrieved in the human brain.

Steps ↓

(a) Multi-store Model:

- This divides memory into 3 parts.

• sensory memory: holds info for a very short time.

• Short term memory (STM): temporary storage (limited capacity).

• long term memory (LTM): permanent storage with large capacity.

Flow: sensor → STM → LTM ~~flow~~

(b) Working Memory Model:

- This is an improved version of STM with multiple components:

- central executive & Controls attention
- phonological loop : processes ~~sound~~ sound / verbal info
- visuospatial sketchpad & processes visual / spatial data.
- Episodic Buffer : Integrates info.

(c) Levels of processing Model :

- Memory depends on depth of processing

- Deep processing → better memory.
- Shallow " → weak "

(d) Connectionist Models :

- Memory stored as patterns in neural n/w.
- Learning happens by adjusting connections (weights).
- Info is distributed, not stored in one place.

(ii) Cognitive Models of Language :

- These models explain how humans understand, produce, & learn language.

(a) Modular Models :

- Language is divided into separate modules
- ex: syntax, semantic etc.

b) Interaction Activation Models:

- Language processing is interactive.

- uses both:
 - Bottom-up (letters $\rightarrow$ words)
 - Top-down (Content $\rightarrow$ meaning)

c) Connectionist (Neural Models):

- Based on neural networks.
- Learn language through experience & patterns.
- No fixed rules; uses probabilities.

d) Dual-Route Model of Reading:

- Lexical route $\rightarrow$ recognize known words.
- Non lexical route $\rightarrow$ read new words using rules.

Integration of memory & language:

- Memory & language are closely connected.
- working memory $\rightarrow$ helps in sentence understanding
- Long term " $\rightarrow$ stores vocabulary & grammar.
- Episodic memory $\rightarrow$ supports content & storytelling

Computational Models of episodic & semantic memory :

⇒ Human memory is mainly divided into :

(i) Episodic memory :- personal experiences (events, time, place)

(ii) Semantic memory :- general knowledge. (facts, concepts, meanings)

⇒ In CC, these helps to stimulate human thinking & improve AI learning & reasoning.

(i) Episodic Memory (Computational Model) :-

- It stores specific experiences with content like :-
 - What happened.
 - When it happened.
 - Where " "

Computational Representation :-

- It is modeled using :-

- Event based data structures
- Time Stamped records.
- Graph based memory w/w.

⇒ It is time dependent & ~~have~~ it is rich in Content.

→ It stores events ~~o~~ sequentially.

Ex

- A chatbot remembers:

- "user asked abt AI notes on April 20".
- "user searched for ~~cc~~ cc topics earlier".

Applications:-

- personal assistants (Siri, Alexa).
- Recommendation systems.

(ii) Semantic Memory (Computational Model):-

- It stores GK independent of time (or) place.

Computational Representations:-

- It is modeled using:-

- Knowledge graphs
- Ontologies.
- Rule based - systems.

⇒ It is fact based.

- It ^{is} stable & have permanant knowledge.

- It is not linked to personal experience.

Expt - A sys knows:

- "Paris is capital of France".
- "Neurons are part of the brain".

Applications :-

- Search engines (Google Knowledge Graph).
- Q/A systems.
- expert systems.

Both Importance in CC :-

- Improves human-like reasoning
- Enhances content awareness
- ~~supports~~ supports ~~learning~~ learning from experience.
- enables personalized AI behavior.

Modeling Psycholinguistics :-

⇒ psycholinguistics is the study of how humans understand, produce, & acquire language.

- In CC, Modeling psycholinguistics means creating computational models that stimulates how the human brain processes language.

⇒ The goal is to :

- understand how humans process language in real time.
- Build sys that can interpret & generate natural language.
- Improve human-Computer interaction.

Components of Language processing :

(i) Language comprehension (Understanding) :

- How humans understand words, sentences & meaning.

ex: understanding a qtn in a chatbot.

(ii) Language production :

- How humans ~~form~~^{form} sentences & speak/write.

ex: ^{AI} generating responses like chatGPT.

(iii) Language Acquisition :

- How humans learn language from childhood.

- used in training AI models using data.

Computational Models in ~~ph~~ Psycholinguistics :

(i) Symbolic Models :

- It is Based on rules & grammar.

- It uses syntax & structure of language.
- ent: parsing sentences using grammar rules.

Advantage: Clear & interpretable.

Disad: Not flexible for complex language.

(ii) connectionist models (Neural N/w)

- It is inspired by brain neurons.
- uses ANN's (Artificial Neural N/w)
- Learns patterns from data.

ent: DL models for NLP.

Advantage: Handles ambiguity & large data.

Limitation: Hard to interpret.

(iii) probabilistic Models

- It is based on probability & statistics.
- It predict likely meanings & (or) next words.

ent: - N-grams.

- Hidden Markov models.

Advantage: - Handles uncertainty.

Limitation: Needs large data.

(iv) Hybrid models

- It combines symbolic + neural + probabilistics approaches.
- Used in modern NLP systems.

Language processing levels in Models:

- (i) phonological level — sounds of speech
- (ii) Morphological " — word formation
- (iii) Syntactic " — sentence structure
- (iv) semantic " — meaning
- (v) pragmatic " — content & intention.

Importance: Helps machine to understand human language naturally.

Applications:

- Improves AI Communication Skills
- Chatbots
- virtual ~~assistance~~ assistants.
- Speech recognition sys.
- Machine translation (google translator)